

T1) Historical Introduction to the In-House Conference

Richard E. Brown

Welcome to the 34th Annual Psychology/Neuroscience In-House Conference. This presentation will introduce you to the people who started the In-House Conference, and who have presented over the past 34 years – a living history of our department! I will end by focusing on faculty who have won awards over the last year.

Title: Disattenuation by Simulation: An estimation and inferential tool for correlation amidst estimable measurement error

Authors: Mike Lawrence

Summary:

Few empirical measurements can be made without some degree of measurement error, and when estimating the correlation between two variables it is well known that this error tends to lead to raw correlation values that underestimate the true correlation. Empirical estimates of measurement error, calculated across repeated measurements of the same entity, can be used to correct these attenuated correlation estimates. A general purpose tool is presented for calculating correlation estimates and corresponding inferential intervals in the context of estimable measurement error. The new method, dubbed disattenuation by simulation (DS), is founded on a simple simulation method that lends a combination of flexibility of application and inferential capacity heretofore unknown in traditional correction procedures. Monte Carlo simulation reveals that DS outperforms traditional approaches to correlation estimation and inference.

Presentation format: Talk

TITLE: Directed Forgetting: Comparing Pictures and Words

AUTHORS: Chelsea Quinlan, Jonathan Fawcett, & Tracy Taylor

ABSTRACT:

The present study used an item method paradigm to examine whether or not a directed forgetting effect would generalize from words to pictures and vice versa. Participants were assigned to one of two study-test conditions: word-picture or picture-word. Either words or pictures were presented one at a time at study

followed by a remember or forget memory instruction. Following the presentation of all the study trials, a yes/no recognition test measured memory performance for both the remember and forget items in the opposite form as study. Results showed that memory performance was significantly greater for to-be-remembered than to-be-forgotten items in both the word-picture and picture-word conditions, which indicates a directed forgetting effect. However, contrary to the predictions of various theoretical views of the picture superiority effect, there was no significant difference in the magnitude of the directed forgetting effect between the two study-test combinations (picture-word vs. word-picture). These findings have important implications for the underlying encoding mechanisms of words and pictures.

PRESENTATION FORMAT: Talk

Development of normal coherent motion perception in kittens requires visual experience in the first 6 weeks of postnatal life

Donald Mitchell and Jan Kennie

Coherent motion perception (CMP) is a complex form of motion that is believed to be processed in primates in the middle temporal cortex (MT) and in cats in PMLS that are both visual cortical area beyond the striate cortex. Here we report that kittens reared in darkness to 6 weeks acquire only rudimentary abilities to perceive CMP despite recovering normal visual acuity. Data from animals deprived for different durations suggest that the critical period during which kittens require visual input is very short, lasting only to 6 weeks. The results suggest that higher visual cortical areas require light early in postnatal life for normal development and are also of interest with respect to the controversial viewpoint of Rosa and colleagues that MT (and possible PMLS in cats) may actually be primary visual cortical areas.

TALK

Memory outcomes after temporal lobe epilepsy surgery: selective amygdalohippocampectomy vs anterior temporal lobe resection

McGlone, Jeannette, Carter, Sherri, and Williams, Jaime

There is no consensus whether cognitive effects such as memory loss are minimized if neurosurgeons remove a smaller than larger region of the temporal lobe. We present retrospective data on 16 patients who underwent selective amygdalohippocampectomy (SAH) versus 48 with anterior temporal lobe (ATL) resections for control of temporal lobe epilepsy. Demographics and medical histories were similar (i.e., age of onset, seizure frequency, medications, side of surgery, seizure outcome). There was no significant between-group difference on objective memory scores, except trends for SAH to have lower post-op Wechsler Paired Associate scores than ATL ($p < .10$). There was no difference on subjective memory ratings using the Memory Observation Questionnaire (MOQ) pre-op or post-op. Within-group ANOVAs (pre to one-year post surgery) showed that objective and subjective scores did not decline significantly after surgery. Interestingly, ATL patients rated memory change more positively than SAH patients (i.e., mean difference on MOQ-SA t score of 8.8, $T_{2,18.2} = 2.1$, $p < 0.06$); a finding that was also seen in the relatives' ratings (MOQ-RA, MD = 9.2, $T_{2,51} = 2.3$, $p < 0.05$; MOQ-RB, MD = 22.4, $T_{2,52} = 2.0$, $p < 0.06$). In this non-randomized sample, subjective memory outcomes favoured the larger ATL than smaller SAH resections.

PRESENTATION FORMAT: Talk ____ Poster X

TIMES/DAY NOT AVAILABLE: April 25 from 11 am until 2 pm (Jeannette)

EQUIPMENT/SOFTWARE REQUIRED: N/A

Predicting subjective memory decline following temporal lobe epilepsy surgery

Carter, Sherri, Williams, Jaime, and McGlone, Jeannette

Memory ratings decline for <10% of temporal lobe epilepsy patients after surgical resections (see Williams, Martin and McGlone, submitted). This retrospective study examined 5/ 64 surgical epilepsy cases who reported significant postoperative memory decline on the Memory Observation Questionnaire (MOQ) determined by reliable change indices. Better pre-surgical memory performance and absence of mesial temporal sclerosis (MTS) were predicted. Cases were matched on age, sex, and full scale IQ (Average range), and then compared on objective memory, mood, and psychosocial adjustment. Wilcoxon tests revealed trends for higher pre-surgical memory performance and greater objective memory loss for those with increased memory complaints post-surgery, although findings did not reach .05 level of significance in this small sample (i.e., WMS Percent Delayed Recall - Stories, $p = .08$ and Verbal Paired Associates, $p = .07$). Contrary to expectations, neuropathology revealed 4/ 5 cases with MTS. Pairs did not differ on depression or psychosocial adjustment yet case review revealed

adverse clinical outcomes for those with more memory complaints. Taken together, these findings concur with Lineweaver et al. (2004), suggesting that for those with normal pre-surgical memory abilities, greater complaints following surgery may coincide with actual memory decline and be a marker of poorer post-surgical outcome.

PRESENTATION FORMAT: Talk ____ Poster X__

TIMES/DAY NOT AVAILABLE: April 24 (Dr. Carter)

EQUIPMENT/SOFTWARE REQUIRED: N/A

The mediational role of reassurance-seeking in the relationship between anxious attachment and hypochondriacal alienation: A preliminary test of the Interpersonal Model of Hypochondriasis

Doucette, S., Sherry, S. B., Stewart, S. H., Hussman, J., Sherry, D. L., & Vincent, N.

Hypochondriasis involves a persistent fear of having a serious illness (American Psychiatric Association, 2000). This common difficulty is debilitating to the sufferer and costly to the healthcare system (Barsky & Borus, 1995). We conducted a preliminary test of the Interpersonal Model of Hypochondriasis (IMH; Noyes et al. 2003). The IMH proposes reassurance-seeking behaviours typical of individuals with hypochondriasis are fuelled by anxious attachment. These reassurance-seeking behaviours lead to instances of conflict with, and feelings of alienation from, other people. Such feelings of alienation are also believed to exacerbate health-related worry among persons with hypochondriasis. We hypothesized that reassurance-seeking would mediate the link between anxious attachment and interpersonal alienation. Fifty-nine undergraduates completed measures of anxious attachment, reassurance-seeking, and interpersonal alienation. Results suggest hypochondriacal reassurance-seeking is a marginally significant partial mediator in the anxious attachment – hypochondriacal alienation connection. This study advances understanding of how hypochondriasis is maintained which may inform future interventions.

PRESENTATION FORMAT: Talk ____ Poster X__

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED: N/A

Impact of Partial Sleep Deprivation on Cortisol Patterns in Young Women

Antonina Omisade¹, Benjamin Rusak^{1,2,3}, and Orfeu Buxton⁴.

Departments of Psychology,¹ Psychiatry² & Pharmacology,³ Dalhousie University, Halifax, NS, Canada.

Harvard Medical School⁴, Harvard University, Boston, MA, USA.

This pilot study examined the effects of partial sleep deprivation on the daily rhythm of salivary cortisol in young women across a range of Body Mass Index (BMI) values. Twelve healthy women (ages 18-25; BMIs 18.3-51.9) spent 3 nights and 2 days at the Chronobiology Laboratory at the QEII HSC. During the first 2 nights (Habituation and Baseline), participants were permitted to sleep for up to 10 h. During the third (Experimental) night they slept for 3 h (05:00-08:00). Saliva samples were collected every 2 h from waking to 21:00 during Baseline and Experimental days. The participants also completed inventories assessing mood and cravings for various food groups at the end of each day. Sleep loss elevated afternoon/evening cortisol levels relative to baseline and attenuated the normal morning peak in salivary cortisol levels. Changes in cortisol patterns were not associated with BMI scores. There was an association between increased evening cortisol levels after sleep deprivation and increased hunger on the Experimental day, elevated cravings for most food groups, and lower mood. There was also an association between elevated evening cortisol on the Experimental day and time since the end of menses. These are the first reported effects of sleep loss on cortisol rhythms in women.

PRESENTATION FORMAT: Talk _x_ Poster _____

TIMES/DAY NOT AVAILABLE: Friday, April 25.

EQUIPMENT/SOFTWARE REQUIRED: No

Title: The perception of biological form and motion: An ERP Study

Authors: Jonathan M. Fawcett and Aaron J. Newman

Summary:

Previous research reporting event-related potentials (ERPs) sensitive to biological motion have not distinguished between stimuli depicting static human forms and motion. In the present study, we addressed the specificity of these

ERP effects to biological *motion* by presenting static point-light displays of upright and inverted human forms that began to move after 400-600 msec. Pre-motion onset, equivalent N170 and N300 amplitudes were observed for upright and inverted human forms; post-motion onset, amplitudes of both N170 and N300 components were larger for upright stimuli. Results suggest that these components are not biological motion specific, but are enhanced by upright biological motion.

Presentation Format: Poster
Times/Day Not Available: N/A
Equipment/Software Required: N/A

Title: Start Forgetting or Stop Remembering? Commonalities Between Directed Forgetting and Stop-Signal Inhibition
Authors: Jonathan M. Fawcett and Tracy L. Taylor

Summary:
By combining an item-method directed forgetting paradigm with a stop-signal task, the current study explored commonalities between directed forgetting and stop-signal inhibition. Following each study phase memory instruction, the current study presented a centralized green circle (Go-Signal) requiring a speeded detection response that was sometimes replaced by a red circle (Stop-Signal) cancelling that response. Although stop-signal reaction times were not affected by the preceding memory instruction, an instruction to Forget did facilitate stopping by delaying responses. These findings provide support for the view that intentional forgetting is an active cognitive process.

Presentation Format: Talk
Times/Day Not Available: *Please*, not a morning session.
Equipment/Software Required: N/A

Disentangling exogenous and endogenous temporal attention
Michael A. Lawrence & Raymond M. Klein

Dalhousie University

The allocation of information processing resources over time, otherwise known as temporal attention, may be achieved by relatively automatic or controlled mechanisms. Loud noises, for example, may automatically elicit attention whilst learned strategies may be used to prepare resource allocation to impending important events based on previously detected temporal contingencies. With analogy to systems of allocation in spatial attention, these theorized mechanisms are termed exogenous and endogenous temporal attention, respectively. Where past research has confounded these theoretically distinct dimensions of temporal attention, the present work seeks to ameliorate this oversight, including a novel application of Rescorla's "truly-random control" procedure.

TITLE: What am I?

AUTHORS: John Barresi

SUMMARY:

"What am I?" is a question that has intrigued thinkers in all cultures at all times. My focus will be answers to this question given by thinkers in the western tradition. Transformations in answering this question have occurred, but the issues remain the same. What is my metaphysical nature? Am I a physical thing, for instance an individual living being, or an immaterial, and immortal substance? Am I even a persistent 'thing' at all? Perhaps I'm a process of some other thing, such as neural activity in the brain? Maybe, instead of being a thing or a material process I'm constituted by being conscious of my self or being self-aware. Or am I merely a concept constructed by the brain or mind but constituted by social processes? Indeed, perhaps, there is no single notion that satisfies all our intuitions about our selves and that I am essentially fragmented in nature.

PRESENTATION FORMAT: Talk ☒X___ Poster _____

TIMES/DAY NOT AVAILABLE: Not available for talk on Friday, March 25.

EQUIPMENT/SOFTWARE REQUIRED: PowerPoint

PRESENTATION TYPE: Talk

Tag, you're it: Tagging as an alternative to Y/N recognition in item-method directed forgetting

Kate Thompson, Jonathan Fawcett and Tracy L. Taylor

In item-method directed forgetting, memory performance is commonly measured using yes/no recognition. Such a task asks the participant whether or not a word was presented during the study phase regardless of memory instruction. The present study proposes another, possibly more valid measure of directed forgetting. Following a standard item-method directed forgetting task, study words were tested using either a standard yes/no task or a tagging task. In the tagging task participants 'tagged' each word as remember, forget, or new. The benefits of the tagging method over a yes/no distinction are discussed.

The Role of Pain in Influencing Children's Recalled Procedural Anxiety
Noel, M. E., Chambers, C. T., McMurtry, C. M., & McGrath, P. J.

Aim of Investigation: Children are required to repeatedly undergo medical procedures over the course of their early lives and many find these experiences to be both painful and frightening. Children's memories of painful medical procedures have important implications for coping during subsequent procedures. Previous research indicates that children's memory for sensory aspects of pain is surprisingly accurate. We examined whether the level of pain children report experiencing immediately after a medical procedure is related to their recalled anxiety 2 weeks following the procedure (above and beyond the anxiety that children initially experience).

Methods: Participants consisted of 27 children (12 males, 15 females) aged 5-10 years ($M = 7.52$, $SD = 1.53$) who had undergone venepuncture. Children completed a measure of pain (Faces Pain Scale-Revised: FPS-R, scores range from 0 to 10) immediately following the procedure and a measure of anxiety (Faces Anxiety Scale: FAS, scores range from 0 to 4) at two time points: immediately and 2 weeks following the procedure. Memory scores were calculated by taking the difference between anxiety ratings at each time point.

Results: After controlling for initial anxiety, the level of pain that children experienced immediately following the procedure was related to their recalled anxiety ($r = .54$, $p < .05$); children who experienced greater pain had higher levels of recalled anxiety. Children who initially reported lower levels of pain (FPS-R = 0-4) tended to accurately estimate their level of procedural anxiety 2 weeks following the procedure. Conversely, children who reported higher levels

of pain (FPS-R = 6-10) immediately following venepuncture tended to report higher levels of procedural anxiety at follow up.

Conclusions: These preliminary findings suggest that children's initial procedural pain experience may be related to how they recall their level of experienced procedural anxiety.

PRESENTATION FORMAT: Talk X Poster

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED: laptop for powerpoint presentation

TITLE: Self-Awareness of Autistic Symptoms, Empathy, and Systemizing in High-Functioning Children and Adolescents with an Autism Spectrum Disorder

AUTHORS: Jillian H. Filliter, MHSA and Shannon A. Johnson, PhD

SUMMARY: Studies of theory of mind in Autism Spectrum Disorders (ASD) typically focus on the ability to perceive and understand others' mental states. It has been proposed that the mechanisms that underlie understanding of others' mental states are similar to those involved in attributing mental states to the self. We compared parent and self-report scores on the Autism Spectrum, Empathizing, and Systemizing Quotients (AQ, EQ, & SQ). Participants were 15 individuals with ASD and 16 age- and IQ-matched controls, ages 9 to 19 years. Comparisons of parent- and self-report scores revealed that ASD participants reported fewer autism symptoms and higher empathy levels relative to their parent, but there was no difference on the SQ. In contrast, parent- and self-reports for the control group did not differ for the AQ and SQ measures, but parents reported more empathizing than participants. These results suggest that those with an ASD have less self-awareness than their typically developing peers.

PRESENTATION FORMAT: A talk is preferred, as this is a requirement of the Proseminar class in which the first author is enrolled at present.

TITLE: PROGRAMMING THE STRESS RESPONSE: INVESTIGATING THE ROLE OF SEX DIFFERENCES IN EARLY HPA AXIS DEVELOPMENT

Amanda D. Green, Tara Perrot-Sinal

Males and females are known to differ in their responses to stressors, and these differences may be related to the observed sex differences in stress-related

diseases. In mammals, differences between the male and female brains are thought to be primarily the result of gonadal hormone secretions, acting during development to 'organize' the wiring of the male and female brain, and in adulthood to 'activate' these differences in neural circuitry. In rodent models, the early postnatal period is a critical time for the sexual differentiation of the brain, and is also a time of significant plasticity in the developing stress system, though little work has directly focused on interactions between the two.

We propose to characterize potential sex differences in the development of several brain structures involved in the modulation of the stress response, as well as investigate some of the potential mechanisms that underly these differences.

PRESENTATION FORMAT: Talk X Poster

TIMES/DAY NOT AVAILABLE: NA

EQUIPMENT/SOFTWARE REQUIRED: KEYNOTE OR PPT

The who, what, when, where and how of knowledge translation: A content analysis of funded CIHR operating grants.

Debbie Johnson Emberly, MSc, Dalhousie University; IWK Health Centre, Ian D. Graham, PhD, VP Knowledge Translation Branch, CIHR & Jacqueline Tetroe, Senior Policy Analyst, Knowledge Translation Branch, CIHR.

Objective: To undertake a content analysis of funded Canadian Institutes of Health Research (CIHR) open access grant proposals to examine the types and budgets of knowledge translation (KT) activities proposed. Methods: A random sample of funded grant proposals (n=50 from funding years 2000/01 and 2005/06) was generated from the total funded operating grants in each of the four CIHR themes. Variables abstracted included characteristics of the grant, investigators, and KT activities. Grants were analyzed for the amount budgeted directed to end of grant KT activities, e.g., conferences, publishing and integrated KT activities, e.g., focus groups with research stakeholders. Results & Discussion: Overwhelmingly, the KT activities identified were end of grant diffusion activities, e.g., conference presentations, publication in peer-reviewed journals. The percentage of grants which identified KT as a specific objective within the proposed research program increased from an average of 10% of all funded applications in 2000/01 to an average of 30% in 2005/06. Innovative KT activities are emerging from the health systems/services and population health themes, e.g., involvement of stakeholders, website dissemination which have the potential to reach more stakeholders for less cost than traditional methods of

dissemination. Implications of these findings are discussed in the context of the mandate of CIHR and the science of KT.

PRESENTATION FORMAT: Talk X Poster

TIMES/DAY NOT AVAILABLE: any

EQUIPMENT/SOFTWARE REQUIRED: none

Personality, Motivation and Patterns of Misuse of Prescription Anxiolytics and Sedatives

Megan E. McLarnon¹, Jessica Meisner¹, Christine Darredeau², and Sean P. Barrett^{1, 2}

¹Department of Psychology, Dalhousie University

²Department of Psychiatry, Dalhousie University

Anxiolytic and sedative medications have important therapeutic uses and are widely prescribed in the treatment of anxiety and insomnia. In addition to their beneficial effects, these substances have psychoactive properties that render them vulnerable to misuse. Despite indications that recreational use of anxiolytics and sedatives is growing, there is a dearth of information available regarding the factors related to their misuse. Existing research has suffered from inconsistency in the operational definition of misuse. Specific personality characteristics and differing motives for use are thought to be related to vulnerability for problematic substance use. The current study investigates anxiolytic and sedative misuse in an adult Canadian sample using a battery of structured interview and self-report measures. Misuse will be operationally defined as the use of the medication for something other than its intended psychiatric purpose. It is expected that distinct clusters of anxiolytic and sedative users will be characterized based on patterns and contexts of use, demographics, personality, psychiatric symptoms, and motivation for use. Identification of the predictors and correlates of anxiolytic and sedative misuse will be essential in understanding this important public health issue.

PRESENTATION FORMAT: Talk x Poster

TIMES/DAY NOT AVAILABLE: none

EQUIPMENT/SOFTWARE REQUIRED: Mac laptop with Keynote '08 software would be great, but I will save my slides in PDF format too just in case.

A Further Investigation of the Relations of Anxiety Sensitivity to Smoking Motives

Susan R. Battista, Sherry H. Stewart, Heather Fulton, Dan Steeves, Christine Darredeau, & Dubravka Gavric

Prior research has shown that anxiety sensitivity (AS) is associated with smoking to reduce negative affect. However, given previous limitations in the measurement of smoking motives, it has yet to be definitively established that AS is specifically related to negative reinforcement smoking motives. The present study examined a sample of 119 smokers attending a tobacco intervention program who completed the Anxiety Sensitivity Index and the Reasons for Smoking Scale (RFS) at pre-treatment. In a principal components analysis of the RFS items, stringent parallel analysis supported a two factor solution (negative and positive reinforcement smoking motives). ASI total scores were significantly positively correlated with both RFS factors in bivariate correlational analyses. In partial correlations, the relation between the ASI and the RFS negative reinforcement factor remained significant when controlling RFS positive reinforcement smoking motives, but the correlation of ASI with RFS positive reinforcement motives was not significant after controlling for RFS negative reinforcement motives. These findings have implications for designing targeted tobacco interventions for high AS smokers.

PRESENTATION FORMAT: Talk ☒ Poster ☐

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED:

How language background affects spelling: The role of first language in spelling English plurals

Keiko Fujise and S. Hélène Deacon

The current study examines whether and how one's first language affects the spelling of the English plural -s. This is examined by comparing the extent to which English learners utilize frequency-based approach (learning through frequent exposure) or rule-based approach (learning through explicit rules) when spelling the English plural -s. The two experimental groups will consist of late learners of English (i.e., exposure to English after age 12) whose first language is French or Japanese. English monolingual control group will also be included.

Whereas French shares the notion of plurality and the writing system with English, Japanese shares neither with English. Consequently, French participants' greater amount of exposure to alphabets and to the notion of plurality may lead them to utilize frequency-based approach in spelling English plurals. On the other hand, Japanese participants may utilize rule-based approach because they do not have the luxury to resort to knowledge built up over time by exposure.

PRESENTATION FORMAT: Talk ☒ Poster ☐

TIMES/DAY NOT AVAILABLE: n/a

EQUIPMENT/SOFTWARE REQUIRED: a Mac with Power Point preferred

Are individual differences in absentmindedness correlated with individual differences in attention?

Yoko Ishigami & Raymond M. Klein

In recent years, absentmindedness in everyday life has been assessed using the Cognitive Failures Questionnaire (CFQ) (Broadbent et al., 1982). We used the CFQ and the Attentional Network Test (ANT) (Fan et al., 2002, Callejas et al., 2005) to examine the relationship between absentmindedness and attention (which has been little explored to date and with inconsistent findings). Using orthogonal subtractions of performance in selected conditions (see Klein, 2002) the ANT measures the efficacy of three isolable components of attention: alerting, orienting, and executive control. Two different versions of the ANT were administered in two experiments with 100 participants in each. Among three networks whose efficiency is assessed by this test, alerting and orienting, but surprisingly not executive control, were correlated with absentmindedness. The opposite correlations between the CFQ and orienting with the two versions of the ANT suggest that their measures of orienting tap separate constructs.

Presentation format: Talk ☐ Poster ☒

Times/day not available: none

Equipment/software required: none

Development of Smiling in Early Infancy: Relations across age and context

Tim Dummer

Infants seem designed to not only orient to but also attract the social elements in their environment that will ultimately aid in constructing knowledge. One effective means for securing adult attention is through the use of facial expressions. Infants display a variety of smiles that differ in terms of their use in social interactions and connection to feelings of enjoyment. Smiles more associated with sleep are termed 'reflexive' because they are believed to occur in the absence of known stimulation. Smiles more associated with alert waking states are termed 'elicited' because they become increasingly more contingent on particular external stimuli. Unfortunately, little is known regarding the function of reflexive smiles and their relationship to elicited smiles. Further, the developmental outcome of individual patterns of smiling during early infancy remains to be discovered. We intend to address the above issues (among others) using a longitudinal design by determining the type, frequency and duration of smiles occurring during active sleep and waking states (both social and non-social contexts).

PRESENTATION FORMAT: Talk

TIMES/DAY NOT AVAILABLE: none (i.e. anytime)

EQUIPMENT/SOFTWARE REQUIRED: I'll just use my laptop (or the Mac book provided)

Better Days Better Nights: Evaluation of a Behavioural Sleep Intervention for Children

Penny Corkum, Trish Pottie, & Patrick McGrath

Objectives: Sleep difficulties are one of the most common concerns reported by parents to family physicians and pediatricians (Owens, et. al., 2002), with approximately 15% to 37% of school-aged children being affected by some type of persistent sleep disturbance (Owens, et. al., 2000). Insufficient sleep can have pervasive negative effects on a child's day-to-day functioning and development (Reid et al., 1999), on the family unit (Kataria, et al., 1987; Richman, 1981) and on child-parent interactions (Stores, 1996). Unfortunately, treatments for sleep problems are not widely implemented (Owens & Mindell, 2002). Although there is research support for the efficacy of behaviourally-based sleep interventions for typically-developing infants and preschool children (Ramchandani, et. al., 2000), there is a paucity of research examining these interventions for elementary school-aged children. Methods: A two-arm, randomized control trial is currently being conducted in which children (5-12 years old), with or without disruptive behaviour disorders, are assigned to a behaviourally-based sleep intervention or

a waitlist control condition. The intervention program consists of five sessions, delivered from a distance using written materials and a non-professional telephone coach. The treatment effectiveness is being evaluated through multiple sleep methods (questionnaires, sleep diaries and actigraphy) and assesses change in sleep disturbance symptoms, daytime behavioural problems, and parental stress. Satisfaction ratings are also gathered from parents. The poster will provide an overview of the study design and treatment, as well as preliminary results from our administrative database regarding treatment effectiveness in terms of impact on sleep disturbances. Results: The preliminary analyses were based on a total of 24 children (12 treatment and 12 control). The treatment group demonstrated significant improvement in the overall sleep disturbances scale ($F(2,21) = 4.20, p = .03$). More specifically, there was a significant change in sleep onset ($F(2,21) = 4.07, p = .03$), but no difference in terms of bedtime resistance ($F(2,21) = 2.30, p = .12$), although the means changed in the expected direction. Parental satisfaction was very high in the treatment group. Conclusions: These results, although preliminary, suggest that the behaviourally-based sleep intervention was effective in improving some sleep patterns. The goal is to have 30 children in each arm who have completed their assignment. After the trial is completed, we will conduct analyses to determine if the results are similar based on actigraphy and whether improved sleep patterns results in improved daytime functioning. Support: Nova Scotia Health Research Foundation, Halifax, Nova Scotia

PRESENTATION FORMAT: Poster X

TIMES/DAY NOT AVAILABLE: April 25 (entire day)

EQUIPMENT/SOFTWARE REQUIRED: N/A

Energy Drinks: What Have You Binge Drinking??

Stephen Price

Research has shown that substance use does not occur in isolation, but frequently in combination with other substances (e.g. Barrett et al., 2006). There is a growing concern that many young people are combining energy drinks (ED) with alcohol (ALC) despite labels warning against this practice. Sixty university students (M 26, F 34, age 17-29, mean = 20.4, SD = 2.4) were recruited who were current or recent ED users. Participants were interviewed to elicit information about their motivations for, and patterns of, ED use, their lifetime substance use and detailed information regarding combined substance/ED use in the seven days preceding the interview. All participants completed self-report

questionnaires (demographics, ADHD checklist, and a personality inventory). Analyses (t-tests) revealed significant differences between those who combine ED/ALC and those who do not, on mean drinks per session, total lifetime substance use, age of first use of ALC and the introversion/helplessness personality scores. Binary logistic regression provided a model for the prediction of ED/ALC co-use.

Presentation: Talk

The Temporal Parameters of Visual Proprioceptive Perception in Infancy

Stephanie Collins and Chris Moore

In humans, differentiating between one's own body movements and those of others is critical. Bahrick and Watson (1985) proposed that the ability to recognize oneself in a mirror and/or live video is aided by the detection of the visual proprioceptive contingency between self and the display. The current study aims to determine at what point children can discriminate between a delayed and non-delayed image of their face. Five to 12-month-old infants will be randomly assigned to one of three delays (1, 2, 10 seconds) and view two computer monitors; one of which displays a live transmission of the infant's face while the other displays a delayed image. It is expected that infants will show a preference for the delayed images, indicating that they can discriminate between the live and delayed feeds. These results will allow us to determine the age at which infants can integrate visual and proprioceptive information.

PRESENTATION FORMAT: Talk __x__

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED: Mac - Keynote (can use my own computer).

Effects of Socializing and Environmental Enrichment on Alzheimer's Mouse Models

Leanne Fraser

The effects of environmental enrichment and group housing are currently being investigated. The study includes transgenic mouse models of Alzheimer's Disease (B6C3-Tg(APP695)3DboTg(PSEN1)5Dbo/J) and their wildtype littermates. Mice are being housed in either an enriched environment (rat cage with tubes, plastic house, nesting material, visual stimuli, and other toys) or an impoverished environment (mouse cage with pine chip bedding only). Within these two environments mice are either singly housed, or housed in groups of 2-4. At 3 weeks of age mice were placed into the experimental housing conditions. Mice will be assessed on a battery of tests at both 6 and 12 months of age. The test battery includes the open field test (exploratory behaviour and anxiety), rotarod test (motor ability), Morris Water Maze (visual learning and memory), olfactory digging task (olfactory learning and memory), and cue and context fear conditioning (learning and memory).

PRESENTATION FORMAT: Talk __X__ Poster ____

TIMES/DAY NOT AVAILABLE: Not available April 25. Would prefer earlier time on April 24th if possible.

EQUIPMENT/SOFTWARE REQUIRED: Digital projector for Power Point presentation

Dangerous Decisions: Examining the Influence of Facial Trustworthiness Assessments on Decisions of Guilt

Leanne ten Brinke, B.Sc. (Hons.) (presenting), Steve Porter, Ph.D., Chantal Gustaw, B.A. (Hons.)

Despite the common assumption that judges and jurors can accurately assess witness credibility, research suggests that they cannot detect deception above the level of chance. According to the Dangerous Decisions Theory (DDT; Porter & ten Brinke, 2008), intuitive evaluations of trustworthiness based on the face greatly influence interpretation of subsequent information about the target. In a courtroom setting, ensuing assessments of evidence may be highly flawed, as evidence is interpreted in a manner consistent with the initial (unreliable) impression of trustworthiness. A recent study by Porter, ten Brinke and Gustaw (2008) presented participants with vignettes describing severe or petty crimes, accompanied by a photograph of the supposed defendant, previously rated as highly trustworthy or untrustworthy in appearance. Participants required less evidence to arrive at a guilty verdict and were less likely to be influenced by exonerating evidence for the untrustworthy-appearing defendants. The current

evidence is supportive of DDT and has implications for legal decision-making practices.

PRESENTATION FORMAT: Talk __X__ Poster ____

TIMES/DAY NOT AVAILABLE: N/A

EQUIPMENT/SOFTWARE REQUIRED: PowerPoint, Laptop (I can take mine if necessary)

The Executive and the Environmentalist: Do Executive Function Skills Help Predict Pro-environmental Behavior?

Meredith Flannery and Sophie Jacques

Recent research into health-related behaviour change suggests that executive function skills may help explain the inconsistency between people's intention to behave in a particular way and their actual behavior (Hall et al., 2006). Like health behaviour, successfully establishing new pro-environmental behaviours may also require executive function skills. Using Ajzen's Theory of Planned Behaviour as a framework, this study examined the extent to which executive function skills helped to explain above and beyond intention the variation in recycling and composting behaviour by undergraduate students. Executive function skills failed to explain significant variation in behavior.

However, performance on the Stop Signal task showed a trend towards significance suggesting that the ability to inhibit a dominant response may be associated with successful behavior. Results are discussed in the context of improving pro-environmental behavior change strategies.

PRESENTATION FORMAT: Talk

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED: Just a computer with PowerPoint

Effects of immune activation on behaviour in the cricket *Gryllus texensis*

Evan R. Fairn and Shelley A. Adamo

As in vertebrates, activating an immune response in insects can result in a suite of behavioural changes called 'sickness behaviour' which are believed to function to increase resistance to infection. We tested whether immune activation causes a change in feeding, agonistic, or shelter-seeking behavior in the field cricket *Gryllus texensis*. Male crickets injected with heat-killed bacteria showed

decreased feeding compared to controls but did not differ from controls in the amount of time spent in shelter. Immune challenged females did not differ from controls in amount of time spent feeding but spent less time in shelter than controls. Immune challenged male crickets generally lost agonistic encounters with control males and immune challenged males were less likely to engage in agonistic behavior.

PRESENTATION FORMAT: Talk x Poster

TIMES/DAY NOT AVAILABLE: none

EQUIPMENT/SOFTWARE REQUIRED: none

Self-generated motives for gambling: Relations to gambling involvement, gambling problems, and co-morbid psychopathology in two population-based samples of gamblers

Sherry H. Stewart, Raymond M. Klein, Sean P. Barrett, and Daniel S. McGrath (presenter)

Motivational models have proven useful in understanding certain addictive behaviours, such as alcohol use; however, they have not been extensively investigated in the gambling area. In the present study, the utility of classifying gamblers into gambling motives groups was examined based on self-generated responses to a question about primary reason(s) for gambling. Secondary data analyses were performed on two gambling prevalence studies. Participant responses were coded in the context of Coopers' (1994) motivational model for alcohol use, adapted for gambling. Categories for primary gambling motives included: coping, enhancement, social, and other. Preliminary results indicate that coping-motivated and enhancement-motivated gamblers are characterized by a greater severity of gambling problems on the CPGI problem gambling index than socially-motivated gamblers. The results also suggest that additional categories may be required to address gamblers primarily motivated by recreation or financial incentives. Motive group differences in demographic variables, psychiatric symptoms, gambling activities, and other substance use will also be discussed.

PRESENTATION FORMAT: Talk Yes Poster

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED:

Exhibitionist flasher uses wing mirrors with fluorescent fusilli for sexual entrapment.

Steve Shaw

A long-legged dolichopodid dipteran from local sea cliffs carries on his wings terminal spots that flash in sunlight over a wide angle, a display aimed to attract the opposite sex. Unusually, the spots behave as mirrors to deflect light incident upon them following the double angle mirror-galvanometer rule, but the wing spot zone is too thin ($1.5\text{ }\mu\text{m}$) to contain any reflector crystals. Instead, broad-band reflecting surfaces are formed from the flattened sides of ~ 5300 ultrathin, colourless hairs ("microtrichia"), extensions of the surface cuticle that jut out at 90° from the wing surface. The puzzle is how their action as mirrors can combine with wide-angle emission. The hairs are actually spiral mirrors, shaped a bit like fusilli pasta and packed like sardines, so an incoming fusillade of light has to undergo multiple randomizing reflections within the tightly packed fusilli. This generates a broadly diffused output beam (measured normal distribution, $\pm 2\text{ SD} \approx 88^\circ$). The microtrichia also fluoresce bright blue in ultraviolet light, prominent in sky-light. While their reflections are mirror-like, microtrichia appear to act as leaky waveguides for this blue fluorescence. Presumably, the flasher-receiver has blue receptors concentrated in part of her retina to take advantage of this parallel, fluorescent visual display.

204 words

- Talk, not poster;
- Afternoon session, preferably grouped with other physiology/neuroanatomy-type talks, if any.
- Powerpoint (Mac not PC)

Impact of Partial Sleep Deprivation on Cortisol Patterns in Young Women

Antonina Omisade¹, Benjamin Rusak^{1,2,3}, and Orfeu Buxton⁴.

This pilot study examined the effects of partial sleep deprivation on the daily rhythm of salivary cortisol in young women across a range of Body Mass Index (BMI) values. Twelve healthy women (ages 18-25; BMIs 18.3-51.9) spent 3 nights and 2 days at the Chronobiology Laboratory at the QEII HSC. During the first 2 nights (Habituation and Baseline), participants were permitted to sleep for up to 10 h. During the third (Experimental) night they slept for 3 h (05:00-08:00). Saliva samples were collected every 2 h from waking to 21:00 during Baseline and Experimental days. The participants also completed inventories assessing mood and cravings for various food groups at the end of each day. Sleep loss elevated afternoon/evening cortisol levels relative to baseline and attenuated the normal morning peak in salivary cortisol levels. Changes in cortisol patterns were not associated with BMI scores. There was an association between increased evening cortisol levels after sleep deprivation and increased hunger on the Experimental day, elevated cravings for most food groups, and lower mood. There was also an association between elevated evening cortisol on the Experimental day and time since the end of menses. These are the first reported effects of sleep loss on cortisol rhythms in women.

PRESENTATION FORMAT: Talk ____ Poster x

TIMES/DAY NOT AVAILABLE: Friday, April 25.

EQUIPMENT/SOFTWARE REQUIRED: N/A

Forget about the flankers

Howells, Fawcett & Taylor

In the item-method of directed forgetting, decreased memory for forget cued words is generally attributed to lack of encoding when memory instructions are presented after each item. The present study examines the possibility of inhibitory mechanisms associated with forget-cued items by adding a flanker task using forget-cued, remember-cued and novel words as flankers in between the study and recognition phase of an item-method directed forgetting task. Flanker trials require a speeded decision about a target word (living or non-living) flanked by words from the same or opposite category. Typically, opposite-category flankers slow reaction times and increase identification errors of the target word. It is hypothesized that, if inhibition is involved in item-method forgetting, the typical flanker effect will be dampened or eliminated when the flanker words had previously been forget-cued, as compared to remember-cued or novel words. Results and implications are discussed.

PRESENTATION FORMAT: Talk X Poster ____

TIMES/DAY NOT AVAILABLE: Unavailable April 25th

EQUIPMENT/SOFTWARE REQUIRED: Laptop & PowerPoint

The Psychophysics of configural face processing

Matheson, H., & McMullen, P.

The present study investigated observer sensitivity to the 2nd order configural relations of features (distances between features) within face and non-face stimuli. Using Just Noticeable Differences (JNDs) we have shown that observers are more sensitive to 2nd order relational information in face stimuli than house stimuli. Contrary to existing theories, inverting the stimuli reduced sensitivity to 2nd order relational information in both faces and houses when both types of stimuli were presented in the same task. However, when only the house stimuli were presented without the face stimuli, inversion failed to reduce sensitivity to 2nd order relational information in houses. Together these results suggest that 2nd order relational processing may not be face-specific, and can be flexibly applied to non-face objects when they are presented in the context of faces.

PRESENTATION FORMAT: Talk YES____ Poster ____

TIMES/DAY NOT AVAILABLE: Saturday is not good.

EQUIPMENT/SOFTWARE REQUIRED: PC with powerpoint

Face perception and the processing of race

Matheson, H., & Newman, A.

Using Event-Related Potentials, researchers have revealed an early negative component--the N170-- that is larger in response to faces than to other classes of object. It is thought that this component reflects processes that encode the structure of the face. Importantly it has been shown that the N170 is not sensitive to the age or gender of the face, suggesting that this type of information is extracted at later stages. However, it is unclear whether this stage is sensitive to the race of the face. In the present experiment, event related potentials were recorded while participants viewed faces of three different races. It was shown that race had no effect on the N170, a finding consistent with the notion that race is extracted from a face image after the basic structure has been encoded.

PRESENTATION FORMAT: Talk YES____ Poster ____

TIMES/DAY NOT AVAILABLE: Saturday is not good.

EQUIPMENT/SOFTWARE REQUIRED: PC with powerpoint

Patterns and Predictors of Tobacco use in Early Psychosis Patients and Controls

Heather G. Fulton, Sean P. Barrett, Sherry H. Stewart, Kimberley Good, Ron Leslie, Alissa Pencer, David Whitehorn and Heather Milliken

Psychosis is a severe disorder affecting more than 1% of the population. Smoking is highly prevalent in this disorder, especially in comparison to the general population. However, little research has examined how individuals with psychosis use tobacco in comparison to controls. Early psychosis patients' patterns of current and lifetime tobacco use were compared to controls. Patients reported their heaviest use of tobacco to be later in life and reported using tobacco more frequently than controls. As different proportions of patients and controls currently used tobacco, subsequent analyses were conducted only with current smokers. Current smokers with psychosis smoked more cigarettes per day and reported greater tobacco dependence levels than current smoker controls. In regression analyses, patient status predicted tobacco dependence levels over and above other variables (e.g., demographics). Results provide insight into how patients with psychosis use tobacco and how to tailor cessation programs to better serve this population.

PRESENTATION FORMAT: Talk X Poster

TIMES/DAY NOT AVAILABLE: none

EQUIPMENT/SOFTWARE REQUIRED: projector for displaying power point slides from my computer.

TITLE: Detecting fMRI activation in white matter: Interhemispheric transfer across the corpus callosum

AUTHORS: E.L. Mazerolle, R.C.N. D'Arcy & S.D. Beyea

SUMMARY: Functional magnetic resonance imaging (fMRI) activation is believed to be restricted to gray matter (GM). Both hemodynamic and neurophysiological evidence support this assertion. In spite of this, fMRI activation is not neatly restricted to GM, and an increasing number of studies are explicitly reporting WM activation. In the current study, we investigated WM activation using 4T fMRI. To elicit WM activation, an interhemispheric transfer task was employed using lateralized cognitive processing. The role of the corpus callosum (CC) in this type of task has been previously established in split brain patients. The analysis revealed CC activation in 14/24 subjects. Of these 14 subjects, four had activation centered in the CC. Interestingly, this activation was consistently localized to the posterior body of the CC. In contrast, CC activation in the remaining 10 subjects was likely attributable to partial volume effects (i.e., contamination by GM signal). Ultimately, characterization of fMRI in WM may open new avenues of clinical

research in which functional assessment of WM is critical.

PRESENTATION FORMAT: Talk X Poster _____

TIMES/DAY NOT AVAILABLE: N/A

EQUIPMENT/SOFTWARE REQUIRED: powerpoint, data projector

TITLE: Detection of fMRI activation in white matter

AUTHORS: Jodie Gawryluk, Kimberly Brewer, Steven Beyea, and Ryan D'Arcy

SUMMARY:

To date, the majority of functional magnetic resonance imaging (fMRI) studies have focused on gray matter because activation is thought to be restricted to these heavily vesseled areas. However, an increasing number of studies are reporting signs of white matter activation. The current study has two main objectives. The first objective is to determine whether it is possible to selectively elicit activation in the corpus callosum with a well known interhemispheric transfer task. The second objective is to evaluate whether differences in MRI contrast mechanisms exist by using a novel method of image acquisition. This technique collects three images (each with differing contrast) that may be compared. Data are being acquired from 10 participants using a 4T MRI with an innovative sequence called asymmetric spin echo spiral. Preliminary results will be presented.

PRESENTATION FORMAT: Talk X Poster _____

TIMES/DAY NOT AVAILABLE: N/A

EQUIPMENT/SOFTWARE REQUIRED: no special equipment required

Title: The structural and physiological analysis of endocytotic organelles and proteins in *Drosophila melanogaster*

Authors: Nancy J Butcher, Tara N Edwards, Birgit Greiner, and Ian A. Meinertzhagen

Summary:

The synaptic vesicle cycle predicts the movement of vesicles through a synaptic terminal so that a reciprocal process of vesicle recovery and neurotransmitter pumping matches the shedding of synaptic vesicles during transmission. The visual system of *Drosophila melanogaster* provides an excellent model for the investigation of the structure and identity of presynaptic organelles and proteins involved in the synaptic vesicle cycle; not only do photoreceptor terminals in *Drosophila* have a regular structure but their synapses are genetically manipulable. In the present study, novel insight into the ultrastructure and identity of proteins and organelles involved in the synaptic vesicle cycle and vesicle endocytosis was provided. New advances in EM tomography enabled the resolution of a filamentous vesicle network within the photoreceptor terminal, and substantiated claims that the T-bar ribbon is a site of vesicle docking and shedding. These reconstructions of synaptic terminals may serve as a basis of comparison for synaptic mutants at a level of detail not previously available using conventional EM methods. Investigation of the two junctional proteins, Crumb (Crb) and Discs Large (Dlg), show that Dlg, but not Crb, is localized to photoreceptor terminals and facilitates the development of the endocytotic organelles known as capitate projections. Unexpectedly, dense core vesicles were found in mutant photoreceptor terminals, potentially uncovering a new function of Dlg in the photoreceptor terminal.

Presentation format: Talk

Times/Day unavailable: Thursday before 1pm

Do Jugglers show Enhanced Processing of Peripheral Visual Stimuli?

Joshua Salmon & Aaron Newman

Action video game players show enhanced performance on a number of attentional tasks. In addition, those who learn a visual spatial language such as American Sign Language show enhanced attention to peripherally presented moving targets.

The current research considered whether or not juggling, another visual spatial task, confers the same benefits as learning sign language or playing video games. Jugglers were matched to non-jugglers of the same age, gender, and handedness and tested on a number of attentional tasks (Multiple Object Tracking, Flanker-Compatibility, and Useful Field of View) as well as having event-related potential recorded during an attention-to-motion task.

Preliminary results (N=12) suggests that juggling may indeed confer some benefits measurable by visuo-spatial tasks. However, the size of the effects

for jugglers may not be as large as those found for video game players and signers, and may be limited to only certain tasks.

Presentation: TALK

Date: No preference.

Equipment Needs: None, beyond a PC computer (preferably).

.....

The Role of Memory in Mathematical Disabilities in Children

Fiona Davidson, BSc, Melissa McGonnell, MSc, Penny Corkum, PhD

Children with learning disabilities commonly experience difficulty with reading and mathematics. While reading disabilities have been the focus of much research, mathematics disabilities have received little attention. There is general agreement that phonological processing is the core deficit associated with reading disabilities (RD). It has been proposed that mathematical calculation is the core deficit in mathematical disabilities (MD). Working memory, particularly as defined by Baddeley and Hitch's multi-component model, has recently been suggested as one cognitive process that may contribute to mathematical difficulties. Participants for the current study were children (ages 7-12) with reading disabilities who completed a battery of measures including the Automated Working Memory Assessment Battery for Children and the math and reading subsets of the Woodcock-Johnson Tests of Achievement-III. The results showed that approximately 50% of children with RD also had MD. Estimated IQ and age were not significantly associated with memory. Results also showed that children with RD+MD are more likely to have difficulty with memory that involves the central executive and visuospatial sketchpad.

PRESENTATION FORMAT:

Talk ____ Poster ✓ ____

TIMES/DAY NOT AVAILABLE:

Thursday April 24th after 1:00 pm

EQUIPMENT/SOFTWARE REQUIRED:

None

.....

TITLE: Neural response of male and female black-capped chickadees (*Poecile atricapillus*) to normal and altered conspecific song.

AUTHORS: Sean P. Roach and Leslie S. Phillmore

SUMMARY:

The purpose of the present study is to establish neural correlates for previous behavioural studies illustrating which particular aspects of the highly stereotyped

song of the black-capped chickadee (*Poecile atricapillus*) are necessary for conspecific recognition by male and female chickadees. Birds will be exposed to synthetic song stimuli of four types corresponding to those used in the behavioural studies: normal songs, and songs with eliminated glissando, altered internote pitch ratio, or both. After sacrifice, levels of ZENK expression in the caudomedial nidopallium (CMM) and caudal medial nidopallium (NCM) will be examined (previous research has established ZENK expression as a reliable indicator of song perception and recognition). It is expected that ZENK levels in these areas will correspond to behavioural responses to normal and altered songs in male and female chickadees. It will be especially interesting to see the neural response of male chickadees to songs with altered internote pitch ratio, as males are able to perceive such differences but do not discriminate based upon them.

PRESENTATION FORMAT: Talk X Poster

TIMES/DAY NOT AVAILABLE: Not available Thursday AM.

EQUIPMENT/SOFTWARE REQUIRED: The available laptop (PC) should be equipped with external speakers.

Adolescent endocrine and behavioural responses to predatory stimuli in a rodent model: Alley cats and hood rats, part II.

Lisa D. Wright, Katherine Muir, Holly McDonald, and Tara S. Perrot-Sinal

Hypothalamic-pituitary-adrenal function and defensive behaviours can be 'programmed' by external factors during early life in rats. Emerging evidence indicates that the adolescent environment is also important in this context, as prefrontal brain regions mature at this time and are responsible for cognitive regulation of stress responding. In order to identify potential mechanisms underlying long-term consequences of adverse adolescent conditions, we conducted a series of experiments designed to examine adolescent responses to stressor exposure. In Experiment 1, we examined responses to two different cat odour stimuli, following repeated adolescent exposures. We also compared adolescent and adult responses to repeated cat odour presentation (Experiment 2), and, finally, we compared adolescent responses to predation threat with responses to a social isolation stressor. Results underscore the importance of the nature of stressor stimuli in determining effects on adolescents, providing support for the hypothesis that adult stress responding will be optimized for predicted environmental conditions.

PRESENTATION FORMAT: Talk X Poster

TIMES/DAY NOT AVAILABLE: none
EQUIPMENT/SOFTWARE REQUIRED: none

Recognition of Emotion in Faces and Voices in High-Functioning Individuals with an Autism Spectrum Disorder

Justin Patterson, B.A. , Shannon Johnson, Ph.D.

It has been proposed that individuals with an Autism Spectrum Disorder (ASD) are impaired at recognizing the emotions exhibited by others. However, previous studies of emotion recognition in ASD have yielded conflicting results. In addition, there has been minimal work addressing the recognition of emotions across modalities in ASD. In two separate, but similar studies, we examined recognition of facial and vocal emotions in a total of 26 children and adolescents diagnosed with Asperger's Disorder and 21 typically developing controls. Participants viewed a series of faces depicting emotional expressions and selected the emotional label for each stimulus. Participants also listened to a series of nonsensical vocalizations expressing emotions and selected the correct emotional label. In Study 1, the Asperger's group was less accurate than controls for the recognition of sad facial expressions and the vocal expressions of both anger and interest. There were no group differences in Study 2.

SUMMARY: Emotion recognition abilities of individuals with Asperger's Disorder

PRESENTATION FORMAT: Poster __X__
TIMES/DAY NOT AVAILABLE: April 24th

Does Perfectionism Impede Research Productivity? A Study of Psychology Professors

Sherry, S. B., Sherry, D. L., Hewitt, P. L., and Flett, G. L.

Amid the "publish or perish" field of academic psychology, personality traits that influence research productivity (i.e., the number of and the impact of journal articles,

book chapters, books, edited books, symposia, and posters) may represent important contributors to career outcomes. In the present study, self-oriented perfectionism (i.e., demanding perfection of oneself) and socially prescribed perfectionism (i.e., perceiving that others are demanding perfection of oneself) were hypothesized to impede research productivity. Psychology professors ($N = 1,258$) completed measures of demographics, personality traits, and research productivity. Results indicated higher levels of self-oriented perfectionism were associated with lower levels of research productivity above and beyond demographics, conscientiousness, and neuroticism. Socially prescribed perfectionism was unrelated to research productivity. The present study suggests psychology professors who demand perfection of themselves struggle to disseminate their research. Publications and presentations generated by psychology professors high in self-oriented perfectionism may be seen as occurring in spite of—and not because of—their perfectionistic behavior.

PRESENTATION FORMAT: Talk X Poster

TIMES/DAY NOT AVAILABLE: An afternoon presentation time is preferred.

EQUIPMENT/SOFTWARE REQUIRED: A PC, PowerPoint, and a projector.

Effects of immune activation on behaviour in the cricket *Gryllus texensis*

Evan R. Fairn and Shelley A. Adamo

As in vertebrates, activating an immune response in insects can result in a suite of behavioural changes called ‘sickness behaviour’ that are believed to decrease the impact of disease on fitness. We tested whether immune activation causes a change in feeding, agonistic (i.e. aggressive), or shelter-seeking behavior in the field cricket *Gryllus texensis*. Male crickets injected with heat-killed bacteria showed decreased feeding compared to controls but did not differ from controls in the amount of time spent in shelter. Immune challenged females did not differ from controls in amount of time spent feeding but spent less time in shelter than controls. Immune challenged male crickets generally lost agonistic encounters with control males and immune challenged males were less likely to engage in agonistic behavior.

PRESENTATION FORMAT: Talk x Poster

TIMES/DAY NOT AVAILABLE: none

EQUIPMENT/SOFTWARE REQUIRED: none

Envisioning the Future: Building a Clinical Neuroscience Program at Dalhousie

Aaron Newman, Shannon Johnson, & Kevin Duffy

Neuroscience research has revolutionized our understanding of relationships between brain, behaviour, thought, and emotion. Tools such as genetic screening and brain imaging will eventually play a central role in the diagnosis and assessment of psychopathology, and evidence from neuroscience already has uses in everyday practice. Although clinical psychologists have a solid understanding of the biological bases of behaviour, a deeper knowledge of the neurophysiological underpinnings of normal and pathological psychological function will be essential in the future. To provide clinicians with this knowledge, and the skill set to use it, training programs must keep pace. Dalhousie, with its exceptional Clinical Psychology program and internationally-recognized strength in neuroscience, is perfectly poised to play a leading role in integrating clinical psychology and neuroscience. In this forum we will outline a 5-year plan to develop a program in Clinical Neuroscience at Dalhousie.

PRESENTATION FORMAT: Talk __X__ Poster ____

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED: Kevlar vests

Interactive and Observational Learning in Infants:

Tristan Neal, Chris Moore, Cherie Collicot

Research in the past 20 years has established infant learning of novel words and social actions in joint attention (see Moore, 2006, for review). Recently, 18 month-olds have shown this learning by observing two people who are socially aloof from the infant interact (Akhtar, 2005; Floor and Akhtar, 2006) and 24 month-olds by observing one socially aloof person perform alone (Nielson, 2006). As of yet, no comparison of learning between these two observational contexts and that of joint attention has been made with the same children. This study compared novel word and action learning between these contexts in a group of twenty-two 14-26 month olds. The results supported previous study of novel action learning, however no evidence for novel word learning was shown. Sample size, children's age, task familiarity and motivation may have influenced the degree of learning demonstrated.

PRESENTATION FORMAT: I would like to do a talk.

TIMES/DAY NOT AVAILABLE: 2:00 pm till night on Thursday.

EQUIPMENT/SOFTWARE REQUIRED: Only a computer/laptop for power-point.

EVERYDAY PAIN INCIDENTS IN CHILDREN AGED 12 TO 36 MONTHS

Kate S. Kalousek, Christine T. Chambers, Patrick J. McGrath, Natalie L. Yanchar, Chris Moore, Carl L. von Baeyer

Aim of Investigation: Everyday bumps and hurts are a common occurrence for children. These minor yet frequent incidents may have considerable influence on children's development of pain coping strategies. While several studies have documented the frequency and nature of everyday pains in preschoolers (3 to 5 years), little is known about everyday pain in younger children. We examined the frequency and nature of everyday pain incidents in a sample of toddlers during active play. Methods: An observational checklist (Dalhousie Everyday Pain Scale - Revised) was used by trained coders to document the nature of minor painful incidents (e.g., body location, cause of hurt, severity of incident, child distress) that took place during the play session. Results: 35 children (18 boys, 17 girls) aged 13-31 months ($M = 22.5$, $SD = 5.2$) and their parents (32 mothers, 3 fathers) were observed at an indoor play facility for up to 4 hours during 2 separate 2 hour play sessions ($N = 17$ for session 1; $N = 18$ for session 2). Analyses will examine how child variables (i.e., sex, age, temperament) influence the nature of the painful incidents as well as parental response to such incidents. The influence of parental reaction (comfort/sympathy, distraction, ignoring, punishment) on the intensity of child distress will also be examined..

PRESENTATION FORMAT: Talk X Poster ____

TIMES/DAY NOT AVAILABLE:

EQUIPMENT/SOFTWARE REQUIRED: I will require a projector for PowerPoint.

I remember your smile: Recognizing facial emotions from a brief acquaintance.

Cindy Hamon-Hill, John Barresi

Does familiarity with a person's emotional responses in particular situations enhance an observer's ability to correctly recognize that person's responses in other situations? Dynamic facial displays of two people were individually paired with images to which they were responses, or paired with other images. For each display, observer's task was to judge whether a person was looking at the

accompanying picture or not. Observers ($N = 60$) were exposed to a pre-test 'acquaintance' period for one of the stimulus faces. Acquaintance improved performance on paired displays previously seen ($p < .001$). Surprisingly, performance on previously unseen expressions did not differ from those for a complete stranger. Familiarity with the person's facial responses did not generalize to unfamiliar situations.

PRESENTATION: Talk
AVAILABLE: Either day.

Epigenetics: The future of psychology

Richard E. Brown

One of the most enduring dichotomies in psychology is the "nature vs. nurture" debate. The human genome project allows us to identify every gene in the human (and mouse, rat, fruit fly, etc) and to understand the contributions of genes to neural and behavioural development. We are also able to identify the physical, social and chemical factors which influence neuro-behavioural development. Epigenetics is the study of how environmental variables regulate gene expression. The action of environmental factors on neurochemical systems and intracellular transduction pathways of neural cells modify the structure of histone proteins in the chromatin of the cell nucleus and these histone proteins regulate gene expression. Thus, epigenetic processes provide the mechanism for environmental regulation of gene expression and the solution to the nature-nurture problem. This talk gives a brief outline of epigenetics and describes epigenetic approaches to the study of behavioural development, aging, learning, memory, social behaviour and neuropsychiatric disorders. Already epigenetic research is changing the way that psychological processes are understood.

The acute effects of exercise or mindfulness therapy on affect, fatigue, and pain in patients with Multiple Chemical Sensitivities

Mark Petter and Chris Blanchard

Multiple Chemical Sensitivities (MCS) is a condition characterized by sensitivity to low levels of food and chemical exposure, which results in a variety of physical symptoms including pain and fatigue. Symptoms may be caused by a

number of common substances such as perfume, detergents, and exhaust. MCS has a large impact on the Nova Scotia health care system, as well as on the quality of life of those with the illness, and research into the effectiveness of treatments is needed.

The Nova Scotia Environmental Health Centre (NSEHC) is a medical facility dedicated to research and treatment of MCS and other environmental illnesses. The NSEHC currently offers mindfulness and graded exercise therapy programs for MCS patients. The purpose of the proposed study is to compare the acute and chronic effects of these two treatment programs on the outcomes of pain, fatigue, mood, and arousal. In doing so, this study intends to delineate the impact of these programs on the MCS patients and examine whether they are the same. A further goal is to determine at what point within each of these programs patients tend to show the most improvement.

The results of the present investigation may improve upon existing MCS treatment programs, and allow for the development of cost-effective, individualized programs. Preliminary results will be presented.

PRESENTATION FORMAT: I would prefer to give a talk.

TIMES/DAY NOT AVAILABLE: I am on the In-House Committee, so whenever is best.

EQUIPMENT/SOFTWARE REQUIRED:

Lack of Local Bias during Selective Attention Global-Local Processing in Autism Spectrum Disorders

Shannon A. Johnson, Leslie M. Blaha, Vanessa A. Bruce, & James T. Townsend

Previous studies of global-local (GL) processing in autism spectrum disorders (ASD; Rinehart et al., 2000; Plaisted et al., 1999) suggest a lack of both global advantage and global interference. However, methods employed in these studies have serious limitations that raise questions about the validity of the results. In the current study, we used traditional cognitive science methods to examine GL processing in ASD during selective attention tasks. Sixteen high-functioning individuals with an ASD and 16 age- and IQ-matched controls completed eight selective attention GL tasks using hierarchical letter and number stimuli.

Repeated measures ANOVAs revealed a main effect of attended dimension with faster response times for the global versus local dimension for both groups on all tasks. Contrary to previous ASD studies, global interference was demonstrated by both groups and neither group demonstrated local interference. Overall, findings indicate typical GL processing in high-functioning ASD participants during selective attention conditions.

Talk

Alcohol Use and Gambling are Associated with Different BAS Components

Roisin M. O'Connor, Sherry H. Stewart, Margo C. Watt, & Kerry MacSwain

Motivational models attribute behaviour to individual differences in the Behavioural Inhibition (BIS) and Activation (BAS) systems. BAS sensitivity has been linked to risk for addictions. Factor analyses suggest that BAS is comprised of drive, fun-seeking, and reward-responsiveness components. Research suggests that the fun-seeking and drive components are most strongly related to alcohol misuse and other externalizing behaviours. We examined the relative contributions of each BIS/BAS subscale to the prediction of 533 undergraduates' self-reported alcohol use, smoking, and gambling behaviour. Factor analysis supported the expected 4 BIS/BAS factors. Controlling for sex and age, hierarchical logistic regressions revealed that only elevated fun-seeking was a significant predictor of any drinking or smoking in the past month, while only low BIS was a significant predictor of any gambling in the past month. Controlling for sex and age, hierarchical linear regressions revealed that only elevated fun-seeking was a significant predictor of alcohol use frequency and quantity, while only elevated drive was a significant predictor of gambling frequency and money spent. Findings suggest that different components of BAS sensitivity contribute to alcohol use and smoking vs. gambling (i.e., the tendency to seek out new potentially rewarding experiences vs. the tendency to strongly pursue appetitive goals, respectively).

PRESENTATION FORMAT: Talk ____ Poster X

TIMES/DAY NOT AVAILABLE: N/A

EQUIPMENT/SOFTWARE REQUIRED: N/A

Characterization of Electrocommunicative Signals of Elephant Nose Fish (Gnathonemus petersii)

Haralambos Lavranos, Leslie Phillmore and Simon Gadbois

Department of Psychology/Neuroscience Institute, Dalhousie University, Halifax, NS B3H 4J1

Elephant nose fish use weak electric signals for both electrolocation and electrocommunication. In this study, we sought to characterize

electrocommunicative signals in two ways: how they change over distance and in response to conspecifics. The signals follow the inverse square law and fall off in maximum, minimum, and root mean square amplitude. The size of the fish was not related to signal power. Small sample size and low signal resolution prevented us from concluding that individual fish discriminate between familiar and non-familiar conspecifics, however there is preliminary evidence that the fish discriminate between an environment with or without conspecifics.

BEHAVIOURAL RESPONSE TO PLAYBACK OF VOCALIZATIONS BY BLACK-CAPPED CHICKADEES (*Poecile atricapillus*)

Laura Robblee, Sean Roach, and Leslie Phillmore

Department of Psychology, Dalhousie University, Halifax, Nova Scotia B3H 4J1

Black-capped chickadees have a well studied and large repertoire of vocalizations used for communication; the production of two of these vocalization types, fee bee song and chick-a-dee call vary with season: song is produced primarily in spring, whereas call is produced year-round (ref). We examined the behavioural responses of black-capped chickadees to playback of three types of vocalizations: (1) conspecific fee bee song, used by males for territorial defense and mate attraction, (2) conspecific chick-a-dee calls, used by both males and females as contact and alarm calls, and (3) heterospecific (Song Sparrow, *Melospiza melodia*) song. We measured the number of vocal responses, the average distance from the sound source, and general activity before, during, and after playback. Overall, there were not many vocal responses to playback of any kind. However, birds appeared to spend more time closer to the speaker during playback of conspecific vocalizations than heterospecific vocalizations, and birds that heard conspecific calls were more active than birds that heard either heterospecific songs or silence. Results are discussed in terms of seasonal influence on behavioural response to vocalizations, and have implications for neural processes underlying perception of vocalizations.

Developing and Testing a New Gambling Motives Questionnaire

Sherry H. Stewart, Tony Toneatto, Martin Zack, Nigel E. Turner, and Valerie V. Grant

Summary: We developed and evaluated a 15-item Gambling Motives Questionnaire (GMQ). In Study 1, principal components analysis of GMQ items completed by 193 gamblers revealed Enhancement (EM), Coping (CM), and Social (SM) Motives factors. Probable pathological gamblers (PPGs) scored

higher across scales than non-pathological gamblers (NPGs) with larger differences for CM and EM. EM predicted gambling behavior, and CM and EM predicted gambling problems. CM predicted gambling problems more strongly in women; EM predicted gambling problems more strongly in men. With gambling behavior controlled, only CM remained associated with gambling problems. 243 problem gamblers, representing six primary gambling activities (e.g., bingo, sports betting), participated in Study 2. GMQ subscales were differentially associated with demographic variables (e.g., CM correlated negatively with being married, working, and income). Gambling motives also varied by gambling activity (e.g., bingo was associated with CM and sports betting with EM). The GMQ appears a promising clinical research tool.

PRESENTATION FORMAT: Talk __x__ Poster ____

TIMES/DAY NOT AVAILABLE: N/A

EQUIPMENT/SOFTWARE REQUIRED: powerpoint

It Is What It Is! (Or Is It, "It Is Where It Is!"?)

Katherine Terauds & Sophie Jacques

The present study aimed to clarify the relative contributions of object and location information to children's awareness of the relation between a symbol and its referent. Three- and 4-year-old children were tested using a version of the DeLoache (1987) model-room search task. Performance on two types of models (congruent vs. conflict) and two types of trials (same-location search vs. different-location search) were compared as a function of whether object properties and object location matched or not. While there was no effect of age, children were able to use object property information but not object location information to search for hidden objects, and when they erred, they appeared to do so by using object-based information.

The potential role of fetal amyloid- β in the pathogenesis of Alzheimer's disease among parous women: Are the kids really driving you crazy?

Jordan Scantlebury, Richard E. Brown, & Tara S. Perrot-Sinal

Epidemiological research suggests that parous women (those with children) are more likely to have a diagnosis of Alzheimer's disease (AD) than nulliparous women (those

without children), perhaps due to a decrease in the average age at disease onset. Parity has not been associated with AD in men. Recent discovery of a moderate increase in amyloid plaque (AP) density with reproductive experience in women, but not men, corroborates these reports. The relationship between parity and AD has been attributed to changes in estrogen and progesterone exposure accompanying pregnancy, but compelling evidence indicates that the role of sex hormones in AD has been exaggerated. The purpose of the proposed research is to investigate a parsimonious alternative mechanism whereby parity may enhance AD pathogenesis, namely, the possibility that amyloid- β ($A\beta$), which is developmentally expressed, is transported from the fetus to the mother during pregnancy. $A\beta$ is the putative pathogenic peptide in AD, and the predominant hypothesis of AD etiology predicts that the presence of fetal $A\beta$ will increase the rate of AP formation and neurodegeneration.

PRESENTATION FORMAT: Talk

AVAILABILITY: Whenever

EXTRA REQUIREMENTS: None

Talk

Title: An habituation account of inhibition of return

Kristie R. Dukewich, Dalhousie University

Inhibition of return (IOR) is an effect whereby reaction times (RTs) to a target presented at a location where a stimulus had recently been presented are increased compared to when a target is presented at a new location. Early reports that habituation is not responsible for the IOR effect are shown to be ill conceived. More importantly, a large portion of the literature can be accounted for using the habituation framework. The current paper reviews some of the support for this theory, and explores some of the predictions that this theory suggests.

Behavioural Consequences of Preferred Direction Coding in Human Motor Cortex

Cowper-Smith, C.D.¹, B.Sc Westwood, D.A.², Ph.D.

To examine the behavioural relevance of the directional tuning preferences for neurons in primary motor cortex (M1), we analyzed the reaction time for consecutive, endogenously cued arm and eye movements (E1: arm-arm; E2: eye-eye). We also explored the existence of a common spatial coordinate system for arm and eye movements by including both types of movements in the same experiment (E3: hand-eye; E4: eye-hand). Participants fixated a central target that was surrounded by 8 equidistant peripheral targets. When the prime cue (a

directional arrow) was displayed at fixation, participants responded with a joystick or eye movement to the indicated target and then returned to centre. The probe cue was delivered 1.5 seconds after the onset of the prime cue, and required a second response (joystick or eye movement); this movement could be to the same target or any of the other targets. The results suggest that arm and eye movements utilize different coordinate systems that do not map onto the known directional tuning curves of neurons in a straightforward manner. The results are broadly consistent with past research on motoric inhibition of return.

TITLE: Using Temporal Order Judgments to Study Spatial Neglect.

AUTHORS: Gail Eskes, Ray Klein, David Shore

SUMMARY:

Spatial neglect after right hemisphere stroke is common and results in difficulties with orienting, detecting or responding to left-sided stimuli. Investigating the underlying mechanisms of neglect can be challenging, due to spontaneous improvement, learned compensatory strategies by patients, and heterogeneity of the neglect syndrome, both within a patient on different tests, as well as between patients. In this talk, we'll describe our approach to studying neglect using temporal order judgments (TOJs) and the preliminary results from a series of studies examining both bottom-up and top-down mechanisms controlling attention in neglect.

Using eye movements to answer basic and applied questions

Raymond Klein et al.

Helmholtz demonstrated that attention and gaze direction could be dissociated and more recently Posner developed the cuing paradigm as a model task for exploring the nature of attention shifts in the absence of shifts in gaze direction. Yet, normally attention and eye movements are closely coordinated. Scholars have, therefore, used oculomotor behavior as a tool for answering basic science questions about cognitive processes and for generating data to shed light on real-world problems. Some of the basic and applied research in the Klein laboratory that uses these strategies will be briefly described.

Preserving Our History with Digital Video: The Snapshot Project, Year 2

By Ron Hoffman and Heather Schellinck

In the winter term last year, students in the Psychology/Neuroscience 3010 class began a project to record on digital media a video "snapshot" of the department

as it exists today, by interviewing faculty members on camera and then producing a short edited version of each interview. Copies of the interviews on DVD have been enthusiastically received by the university archivist, and may some day be available in a departmental museum. This year's 3010 class completed an additional nine interviews. In this presentation, we will briefly describe the project and the training given the students, and show the resulting material that will go into the departmental and university archives.